

≡ Example

Suppose we have this:

$$\begin{cases} \dot{x} = \begin{pmatrix} 1 & 0 \\ 1 & 2 \end{pmatrix} x \\ x(0) \text{ known} \end{cases}$$

What is the solution?

The goal is to find a transformation matrix $T = [u_1, u_2]$, with u_1 and u_2 being the eigenvectors.

Step 1 is to find the eigenvalues of A , and we have to solve the characteristic equation: $\det(A - \lambda I) = 0$.

$$A - \lambda I = \begin{pmatrix} 1 & 0 \\ 1 & 2 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 - \lambda & 0 \\ 1 & 2 - \lambda \end{pmatrix}$$

$$|A - \lambda I| = (1 - \lambda)(2 - \lambda) - 1 \cdot 0 = (1 - \lambda)(2 - \lambda) = 0$$

Thus, $\lambda_1 = 1$ and $\lambda_2 = 2$. These are the eigenvalues, which are real and distinct.

For step 2, we need to find the associated eigenvector for every eigenvalue.

Let's start from $\lambda_1 = 1$, we need to find u_1 . It can be found by solving $Au_1 = \lambda u_1$, which we write in the following form: $(A - \lambda_1 I)u_1 = 0$. The first part is known, and we have to find u_1 . Let's define $u_1 = \begin{pmatrix} a \\ b \end{pmatrix}$.

$$\left[\begin{pmatrix} 1 & 0 \\ 1 & 2 \end{pmatrix} - \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \right] \begin{bmatrix} a \\ b \end{bmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix} \cdot \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Thus, the resulting equations will be:

$$\begin{cases} 0 = 0 \\ a + b = 0 \end{cases}$$

Then, we can pick values of a and b , like $a = 1$ and $b = -1$. So we get $u_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$.

So, let's perform a check:

$$\begin{pmatrix} 1 & 0 \\ 1 & 2 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix} = 1 \cdot \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

So, we've found the eigenvector associated to $\lambda_1 = 1$.

Now, let's find u_2 . We'll have to solve this equation:

$$(A - \lambda_2 I) \cdot \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \rightarrow A - \lambda_2 I = \begin{pmatrix} 1 & 0 \\ 1 & 2 \end{pmatrix} - \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$$

$$\begin{pmatrix} -1 & 0 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

We get the following system:

$$\begin{cases} -a = 0 \\ a = 0 \end{cases}$$

Thus, b can be any number different from zero (to not have something trivial).

Let's pick $u_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$. This is the eigenvector associated to λ_2 .

For step 3, we have to compute $T = [u_1, u_2]$ and $T^{-1} = \begin{bmatrix} v_1^T \\ v_2^T \end{bmatrix}$.

To compute T , we simply put together the two vectors:

$$T = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix} \rightarrow T^{-1} = \frac{\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}}{\det(\cdot)} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$$

We can check that $T \cdot T^{-1} = I$.

$$T \cdot T^{-1} = I \rightarrow \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

For step 4, we need to write the free response of the system.

$$x_{free}(t) = e^{\lambda_1 t} \cdot u_1 \cdot v_1^T \cdot x(0) + e^{\lambda_2 t} \cdot u_2 \cdot v_2^T \cdot x(0) =$$

$$= e^t \cdot \begin{pmatrix} 1 \\ -1 \end{pmatrix} \cdot (1 \ 0) \cdot x(0) + e^{2t} \cdot \begin{pmatrix} 0 \\ 1 \end{pmatrix} \cdot (1 \ 1) \cdot x(0)$$

We can gather then things:

$$\left[e^t \cdot \begin{pmatrix} 1 & 0 \\ -1 & 0 \end{pmatrix} + e^{2t} \cdot \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix} \right] x(0) = \begin{pmatrix} e^t & 0 \\ -e^t + e^{2t} & e^{2t} \end{pmatrix} \cdot x(0)$$

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Let's now introduce the concept of Stability.

It's maybe the most important property of a control system. We may think of stability as a property of a system, rather it is a property of the equilibrium points of a system.

What are equilibrium points?

Let's consider a generic system, $\dot{x} = f(x, u)$. We are not considering the control as of now, but only the autonomous equation. Why do we do this? Imagine we have a system of which we can measure the entire state x . The controller $C(x)$ will take x , and output u , which will be a function of the state $u = C(x)$.

The equations of a system under control are given by:

$$\dot{x} = f(x, C(x)) = \tilde{f}(x)$$

We now call equilibrium points x^{eq} . We have that x^{eq} is an equilibrium $\iff$

$$f(x^{eq}) = 0$$

To find eq points, we need to solve $f(x) = 0$.

Thus, we have that $\dot{x} = f(x^{eq}) = 0$.

When the system is in an equilibrium point, it does not move, and remains there forever.

An obvious example is given by a simple pendulum, to which we're not applying any force. We can have a family of equilibrium when the pendulum is perpendicular to the "wall".

In general, a system can have:

- no equilibrium points,
- one equilibrium,
- some equilibrium
- infinite many equilibria.

Consider a linear system $\dot{x} = Ax$. In general, the rank of A is full ($|A| \neq 0$), so we have a system of independent equations. So, the equation $Ax = 0$ has only the trivial solution $x^{eq} = 0$.

Now, what is stability? It's a property of equilibrium points. In practice, x^{eq} is:

- (simply) stable if after an initial disturbance the trajectory of the system remains close to the equilibrium.

- asymptotic stability (AS): after an initial disturbance the trajectory of the system go back to the equilibrium.
- unstable: not stable.

Now, how to check if a system is stable or not? It's fundamental for safety, and we want to do it for several reasons. If we have a process that is not stable, the first objective of any controller design is to stabilise the system, then comes everything after.

This is called the "stabilisation problem": we have an unstable process, and we add a controller to make the total new process stable. In fact, by adding a controller we're also changing the properties, creating a new system.

Also, imagine we have a stable system, and we add a controller to make our plant faster. When we add a controller to a process, we have to check that the resulting total system is stable as well.

Let's take a look now at stability criteria. We have:

- criteria for linear systems
- criteria for non linear systems. These are known as Lyapunov criterion, which we'll take a look at when discussing switching attacks and load altering attacks.

Let's focus on stability criterion for linear systems.

By having this:

$$\begin{cases} \dot{x} = Ax \\ x^{eq} = 0 \end{cases} \longrightarrow \text{for linear systems, we can compute } x(t)_{free}$$

$$x(t)_{free} = e^{\lambda_1 t} u_1 v_1^T x(0) + \dots + e^{\lambda_n t} u_n v_n^T x(0)$$

This is the solution when λ_i are real and distinct.

Criterion for simple stability: What is simple stability? It means that when $x(0) \neq 0$, the trajectories $x(t)_{free}$ remain limited (around the origin). When this happens, we need to look at the exponentials. If even one of these exponentials is positive, the state goes away, and thus the equilibrium is unstable. For the state to remain close, all λ_i must be negative, or zero.

If we are in a simple case of λ_i real and distinct, we have that:

- S (simply stable) $\iff \lambda_i \leq 0 \ \forall i = 1, 2, \dots, n$
- AS $\iff \lambda_i < 0 \ \forall i = 1, \dots, n$

- U (unstable) $\iff \exists i: \lambda_i > 0$.

Case two is given by eigenvalues λ_i which are distinct, but some are complex. We have pseudoperiodic modes. What makes the difference is the real part of the complex eigenvalue. To have a convergence, we need to have $\text{Re}(\lambda_i) < 0$.

If $\text{Re}(\lambda_i) = 0$ it's still stable, however if $\text{Re}(\lambda_i) > 0$ then we have instability.

Case three: some λ_i are coincident. This can give rise to polynomial modes, like $e^{\lambda_i t} \cdot t^*$. t^* is dangerous, as if we have $e^{0t} = 1$ and constant, however if we multiply it by time it's diverging.

So, if we have multiple eigenvalues in 0 (or with real part equal to zero), we could have unstable polynomial modes.

In conclusion, we can plot the complex plane to give a final criteria for linear systems. If the system has even eigenvalue with positive real part, then it's unstable.

So, for instability, we must have at least one λ_i with positive real part ($\text{Re}(\lambda_i) > 0$).

For AS , a linear system is $AS \iff$ all λ_i have negative real parts ($\text{Re}(\lambda_i) < 0$).

Finally, the criterion for S , is a system is $S \iff$ all λ_i have negative real part, or at most 0 real part. Here we have the important warning: if the λ_i with 0 real part are distinct, then system is stable. Though, some λ_i with $\text{Re}(\lambda_i) = 0$ could be coincident, and we'll have to check more in detail.

A matrix tells us everything about a linear system state evolution.

Let's write for a second the full solution:

$$x(t) = e^{At}x(0) + \int_0^t e^{A(t-\tau)}Bu(\tau) d\tau$$

$$y(t) = Ce^{At}x(0) + \int_0^t Ce^{At}Bu(\tau) d\tau + Du(t)$$

So far we've focused on the spectral formula of e^{At} , how to decompose the matrix, in different modes. By putting the extended formula, we can realise this: the free evolution of the system will contain all the modes of the system. In the forced evolution, we have instead the multiplication with B and the inputs. The product of $v_i^T B$ could be zero, which is not good for controllability of the system, blocking us from pushing the system towards one direction. In fact, there will be some parts of the system we cannot influence with our control.

Now, what is the stability criterion for discrete time linear systems?

We have something like this:

$$\begin{cases} x(k+1) = Ax(k) \\ x(0) \text{ known} \end{cases}$$

What is the solution? It's given by $x(k) = A^k x(0)$. In the solution, we have A^k instead of e^{At} found in $x(t)$ - continuous time.

Also here we'll diagonalize the system, allowing us to compute the power easily.

So, in continuous time e^{At} gives rise to exponential modes.

In discrete time, A^k give rise to λ_i^k . So, to determine the stability criteria, we have to look at λ_i^k .

We can sketch the complex graph. The boundary is not given by the "ordinata", rather by the unit circle, having radius of 1. If all eigenvalues are inside, we have asymptotic stability. If some are on the circle, it's stability. If some are outside the circle, then we have instability.

The essence of control changes the eigenvalues of the system.

Assume that we have this system:

$$\begin{cases} \dot{x} = Ax + Bu \\ y = x \end{cases}$$

One simple and powerful control is the proportional control, which is proportional to what we measure, so given by

$$u = K \cdot x$$

The output is a static state (?) feedback, and the controller is static.

By replacing the controller, we have the following system equations:

$$\begin{cases} \dot{x} = (A + Bk)x \\ y = k \end{cases}$$



What reasons can we have for control?

1. Stabilisation.

2. Performance. Eigenvalues also tells us how fast is the system, how fast the trajectory $x(t)$ moves.

Now, let's take a look at linearization of nonlinear models.
Let's take the pendulum as an example, with θ the angle.



We have the following physical equation modelling the system:

$$\ddot{\theta} + \frac{g}{L} \sin(\theta) = \frac{1}{mL^2} \cdot \tau$$

The goal is to control θ and $\dot{\theta}$ by acting on τ .

Let's put it in state space form: $x_1 = \theta$, $x_2 = \dot{\theta}$.

We have then:

$$\begin{cases} \dot{x}_1 = \dot{\theta} = x_2 \\ \dot{x}_2 = \ddot{\theta} = -\frac{g}{L} \sin(\theta) + \frac{1}{mL^2} \tau \end{cases}$$

Finally, we have the state equations:

$$\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = -\frac{g}{L} \sin(x_1) + \frac{1}{mL^2} u \end{cases}$$

For y , let's say we want to control the position and velocity, so $y = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} x$.

This is an example of a non linear system, which can be written in general in this way:

$$\begin{cases} \dot{x} = f(x, u) \\ y = g(x, u) \end{cases}$$

For non linear systems, we can approximate the behaviour of them close to an equilibrium point by linearizing the model.

The intuition is this: imagine we have one state variable on abscissa, and $\dot{x} = f(x)$ on the ordinates. With linearization, we fix a point over the graph, and there we

approximate the curved function with the tangent. The point we approximate to is the equilibrium point x^{eq} . So:

$$\dot{x} = f(x, u) \approx f(x^{eq}, u^{eq}) + \left. \frac{df(x, u)}{dx} \right|_{(x^{eq}, u^{eq})} \cdot (x - x^{eq}) + \left. \frac{df(x, u)}{du} \right|_{(x^{eq}, u^{eq})} \cdot (u - u^{eq}) + \dots$$

We neglect higher order terms.

We have $\left. \frac{df(x, u)}{dx} \right|_{(x^{eq}, u^{eq})}$, $B = \left. \frac{df(x, u)}{du} \right|_{(x^{eq}, u^{eq})}$, ...

$$\Delta \dot{x} = (\dot{x} - \dot{x}^{eq}) = \dot{x} = \cancel{f(x^{eq}, u^{eq})} + A\Delta x + B\Delta u$$

Which is good for

...

Same for the output, we can write it as:

$$y = g(x, u) \approx g(x^{eq}, u^{eq}) + \left. \frac{dg}{dx} \right|_{(x^{eq}, u^{eq})} \cdot \Delta x + \left. \frac{dg}{du} \right|_{(x^{eq}, u^{eq})} \cdot \Delta u$$

We can then write:

$$y - \underbrace{g(x^{eq}, u^{eq})}_{y^{eq}} = C\Delta x + D\Delta u$$

For the pendulum:

$$(x^{eq} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, u^{eq} = 0)$$

And now, let's try to compute this:

$$\left. \frac{df}{dx} \right|_{(x^{eq}, u^{eq})} = \begin{pmatrix} \cdot & \cdot \\ \cdot & \cdot \end{pmatrix}$$

We have four elements. The first is zero (*I didn't get why*), the second is 1, the third one is $-\frac{g}{L}\cos(x_1)$, and the fourth is zero. We need to compute it with $\begin{pmatrix} 0 \\ 0 \end{pmatrix}, 0$, so

$$\underbrace{\begin{pmatrix} 0 & 1 \\ -\frac{g}{L} \cdot 1 & 0 \end{pmatrix}}_A \begin{pmatrix} 0 \\ 0 \end{pmatrix}, 0$$